ACROPOLIS
Enlightening wisdom

# Presentation on
# NetAgent AI-Powered Network Security System


**Guided by:**

Prof. Nitin Kulkarni
Mr. Mehul Purohit

**Presented by:**

Ram Neekhra (0827IT231110)
Riya Yadav (0827IT231112)
Sarthak Purohit (0827IT231119)
Sumit Singh (0827IT231137)

# Problem Statement

## The Cognitive Gap & Log Fatigue

• **Unstructured Data Overload:** Existing security stacks fail to autonomously convert high-volume raw telemetry into structured context, overwhelming analysts with millions of events per second.

• **Critical Delays:** Mean Time to Action (MTTA) typically spans 2 to 4 hours as human analysts manually sift through and correlate logs from disparate sources.

• **Expanded Vulnerability Window:** Protracted delays allow sophisticated threats to persist and escalate before a response is mounted, causing significant damage.

**Objective:** Reduce MTTA from hours to under 10 minutes by synthesizing a high-fidelity report detailing WHAT happened, WHY it succeeded, and HOW to respond

# Existing Systems & Limitations

**Nmap (Gordon Lyon)**

Features: Active scanning, OS fingerprinting, network mapping

Limitations: Provides raw data only; lacks context and reasoning; requires manual analysis

**Scapy (Philippe Biondi)**

Features: Deep packet inspection, protocol manipulation

Limitations: Highly manual tool; requires deep protocol expertise; not autonomous

**Traditional IDS (Snort, Suricata)**

Features: Detects known threats using rigid attack signatures

Limitations: Fundamentally reactive; blind to zero-day exploits and polymorphic malware

**Current SIEM Systems**

Features: Log aggregation, alert generation

Limitations: Fail at real-time context generation; overwhelming human analysts with massive quantities of unstructured data

# System Architecture



**1. The Probe (Raspberry Pi)**

- Frontline network sensor performing targeted Deep Packet Inspection (DPI)
- Active reconnaissance using Nmap and Nikto for network mapping
- Extracts and curates only essential metadata (connection logs, protocols, IOCs)
- Eliminates bulk log data to reduce bandwidth and storage

**2. The Brain (Edge Compute Node)**

- Powered by quantized LLM (Llama 3) via Ollama runtime
- Implements ReAct (Reasoning and Acting) framework for intelligent analysis
- Uses LangChain/CrewAI for orchestrating Chain-of-Thought investigations
- 100% local processing ensures data sovereignty and privacy

**3. Recursive Validation Loop**

- AI acts as master orchestrator executing specialized security tools
- Analyzes tool outputs and formulates new hypotheses iteratively
- Validates findings before alerting users to minimize false positives
- Continuous self-correcting mechanism ensures thorough threat vetting

# Technologies Used



**Hardware Infrastructure**:

• Raspberry Pi 4/5: Acts as frontline network probe for DPI and active scanning

• Edge Node (Laptop/PC): Hosts the quantized LLM (Ollama) for AI reasoning

**Artificial Intelligence & Machine Learning:**

• Ollama: Local LLM runtime engine for offline reasoning without internet dependency

• Llama 3: Quantized Large Language Model for resource-efficient edge deployment

• LangChain / CrewAI: Agentic frameworks managing ReAct Chain-of-Thought loops

**Development & Integration:**

• Python 3.10+: Primary language for tool integration and agent orchestration logic

• Node.js: Automation and script execution for secure tool coordination

**Security Tools Stack (Orchestrated by AI):**

• Nmap: Network reconnaissance and OS fingerprinting for topology discovery

• Nikto: Web server vulnerability scanning for application-layer threats

• Scapy: Custom packet crafting for deep protocol analysis and anomaly detection

• Searchsploit: Vulnerability database integration for contextual threat correlation

# Results & Key Findings



**Performance Metrics:**

✓ MTTA Reduction: <10 minutes (vs. 2-4 hours with manual analysis)

✓ Automation Level: 100% autonomous tool orchestration without human intervention

✓ False Positive Reduction: Minimal through recursive validation mechanism

✓ Data Privacy: 100% local processing with zero data exfiltration

**Threat Detection Performance:**

✓ Successfully detected and analyzed port-scanning campaigns across network segments

✓ Identified SQL injection and directory traversal attacks with high confidence

✓ Detected Command and Control (C2) beacon communications and lateral movement

✓ AI orchestrator demonstrated 100% autonomous execution of complex tool chains

✓ Generates actionable intelligence with detailed WHAT, WHY, and HOW analysis

✓ Recursive validation loop effectively minimizes false positives and alert fatigue

**System Capabilities:**

✓ Handles millions of network events per second without data loss

✓ Processes threat events completely offline without cloud dependencies

✓ Maintains full audit trail and evidence correlation for incident response

# Conclusions

**1. Hybrid Architecture Innovation:**

Successfully developed a hybrid agentic architecture that cleanly separates IoT sensing (Raspberry Pi) from AI reasoning (Edge Node), enabling scalability and resource optimization.

**2. Edge AI Autonomy:**

Demonstrated that locally-hosted quantized LLMs can autonomously orchestrate complex, multi-stage security tools without requiring human intervention or external APIs.

**3. Recursive Validation Loop:**

Introduced a novel recursive validation mechanism that drastically reduces false positives by verifying hypotheses before surfacing alerts to security teams.

**4. Dramatic MTTA Improvement:**

Achieved < 10 minutes Mean Time to Action, transforming response capabilities from hours to near real-time, enabling proactive threat mitigation.

**5. Data Sovereignty Guarantee:**

Ensured 100% data sovereignty by implementing all AI reasoning, inference, and packet inspection locally on edge hardware, maintaining strict compliance with global data privacy regulations (GDPR, HIPAA, etc.).

**6. Actionable Intelligence Framework**:

Created a structured intelligence framework that converts thousands of raw security alerts into concise, human-readable reports with clear remediation steps.

**7. Enterprise Viability:**

Proved the feasibility and effectiveness of Edge AI for enterprise-grade cybersecurity applications without requiring centralized SOCs or expensive skilled analysts.

# Future Scope



**Short-term Enhancements (3-6 months):**

• Multi-Agent Distributed Architecture: Deploy multiple NetAgent instances across network segments for distributed threat detection and coordinated automated response

• Advanced Threat Intelligence Integration: Integrate with OSINT platforms and threat feeds for enhanced contextual analysis

• Improved LLM Fine-tuning: Develop domain-specific models trained on enterprise security data

**Medium-term Expansion (6-12 months):**

• Cloud-Edge Hybrid Deployment: Extend to hybrid cloud-edge environments while maintaining data sovereignty and privacy guarantees

• Fully Autonomous Incident Response: Implement automated response mechanisms with human-in-the-loop approval workflows for critical actions

• Regulatory Compliance Automation: Automatically generate compliance reports for GDPR, HIPAA, PCI-DSS auditing and regulatory requirements

**Long-term Vision (12+ months):**

• Enterprise SOC Replacement: Transform into a comprehensive Security Operations Center replacement with advanced threat hunting and predictive capabilities

• Industry Standardization: Establish NetAgent as an open standard for edge-based AI security

• Global Security Mesh: Enable federated deployment across organizations for collaborative threat intelligence and response.

# Thank You

## Queries ?